

MEMOLOGS

Part A — Persona Prediction & Trust-Aware Multimodal ML System

Technical Report · AI/ML Engineer Take-Home

Objective

Given an advertisement creative, predict the most likely target persona(s), retrieve supporting evidence, detect near-duplicate creatives, prevent duplicate leakage during evaluation, and decide whether the system should suggest a persona, suggest with caution, or abstain.

Dataset	2,882 creatives + 2,882 images
Human-labelled set	600 creatives
Allowed personas	8
Duplicate evaluation	120 held-out pairs (60 duplicate / 60 non-duplicate)
Final embedding	CLIP ViT-B/32, 512-D fused multimodal representation
Final fusion	30% image + 70% text
Final classifier	One-vs-Rest Logistic Regression
Final test Macro-F1	61.10%
Final test Micro-F1	64.89%
Final test Hamming loss	8.97%
Final subset accuracy	51.09%

This report is based on the project notes and experiment record supplied for the assignment. It preserves the actual experimental progression and final measured results.

1. Executive Summary

Part A was implemented as a multimodal, leakage-aware, multi-label persona prediction system. The system starts from advertisement images and ad text, converts both modalities into CLIP embeddings, fuses them into a normalized representation, and uses the representation for duplicate detection, evidence retrieval, and persona prediction.

A key design requirement was to compare two persona approaches rather than reporting a single model: (1) similarity-based k-nearest-neighbour label propagation and (2) a classifier trained on the multimodal embeddings. The final classifier was One-vs-Rest Logistic Regression, selected using the development set.

The experimentation progressively improved methodological quality. Duplicate detection was converted into complete-linkage clusters to prevent near-duplicate leakage; train/dev/test decisions were made at the cluster level; persona thresholds and hyperparameters were tuned only on the development set; multiple image/text fusion ratios were tested; and CLIP ViT-B/32 was compared with SigLIP as an alternative vision-language representation.

Final held-out result: with 30% image + 70% text fusion, One-vs-Rest Logistic Regression using $C=30$ and a 0.60 persona decision threshold achieved 61.10% Macro-F1, 64.89% Micro-F1, 64.13% Weighted-F1, 8.97% Hamming loss, and 51.09% exact subset accuracy. The corresponding kNN baseline achieved 49.32% Macro-F1, 55.10% Micro-F1, and 11.96% Hamming loss.

The final inference layer adds abstention, evidence retrieval, schema validation, and an evidence-constrained LLM rationale. This turns the system from a raw classifier into a trust-aware prediction pipeline.

Key takeaway

The strongest result came from using a pretrained multimodal representation and a lightweight classifier rather than training a large model from scratch. The small labelled set is used to learn persona boundaries over an already semantic embedding space.

2. Problem Definition & Assignment Requirements

The Part A task asks the system to take a creative identifier such as c_0412 and produce a structured output containing a decision, one or more persona predictions with scores, supporting evidence, a rationale, and the source of the rationale.

Required output concept

```
{"creative_id":"c_0412","decision":"suggest","personas":[{"name":"Traveler & Leisure","score":0.81}],"evidence":[{"creative_id":"c_0077","similarity":0.93,"near_duplicate":true}],"rationale":"...","rationale_source":"llm"}
```

Core requirements

- Create embeddings using both image and text.
- Evaluate duplicate / near-duplicate detection with precision and recall-oriented metrics.
- Use the 600 labelled creatives to predict personas for the remaining unlabelled creatives.
- Compare at least two approaches: kNN label propagation and a classifier trained on embeddings.
- Prevent duplicate creatives from leaking between train and evaluation sets.
- Support multi-label persona prediction.
- Implement abstention rather than forcing a prediction on every creative.
- Provide retrieved evidence and an LLM-generated rationale constrained to that evidence.
- Validate LLM output and provide a deterministic fallback.

3. Data Understanding

3.1 Dataset inventory

Asset	Size / role
creatives.parquet	2,882 advertisements; creative_id, ad_size, ad_text, image_path, launched_at
images/	2,882 actual ad images across multiple banner dimensions
confirmed_personas.csv	600 human-confirmed labelled creatives
personas.yaml	8 allowed persona definitions
dup_pairs_eval.csv	120 duplicate-evaluation pairs; 60 positive, 60 negative
geo_panel.csv	Present in the project root for Part B; not used for the Part A persona model

3.2 Creative and image inspection

The main creative table contains no missing values in the inspected columns. The image set contains eight observed dimensions, making preprocessing important because visually equivalent campaign creatives can appear in different ad formats.

Image dimensions	Count
300 × 250	1,090
728 × 90	935
300 × 600	321
970 × 250	240
970 × 90	109
160 × 600	100
320 × 50	50
336 × 280	37

3.3 Persona labels

The persona task is multi-label: a creative may belong to more than one persona. Labels are stored using the pipe delimiter (|). The inspected label distribution is strongly imbalanced.

Persona	Confirmed creatives
Traveler & Leisure	146
Health & Wellness Seeker	113
Tech Enthusiast	99
Small Business & B2B	93
Student & Learner	83
Homeowner	70
Retiree & Senior	16
Parent & Family	13

The low counts for Retiree & Senior and Parent & Family make Macro-F1 particularly important: an aggregate metric dominated by frequent personas could conceal weak performance on minority personas.

4. End-to-End System Architecture

The final Part A pipeline is organized as a sequence of representation, duplicate-control, learning, trust, and explanation stages.

Stage	Function
1. Data ingestion	Load creatives, images, confirmed labels, and persona definitions.
2. Multimodal embeddings	Encode image and text with CLIP and fuse normalized vectors.
3. Near-duplicate detection	Compute cosine similarity and identify high-similarity creative pairs.
4. Duplicate clustering	Group near-duplicates with complete-linkage hierarchical clustering.
5. Leakage-safe split	Assign whole duplicate clusters to train/dev/test.
6. Persona models	Compare kNN label propagation against One-vs-Rest Logistic Regression.
7. Abstention + evidence	Determine trust state and retrieve labelled supporting examples.
8. LLM rationale	Generate a concise evidence-constrained explanation; validate and fall back deterministically.
9. CLI JSON	Expose the complete prediction through a reproducible command-line interface.

High-level flow

```

Image ████ CLIP image embedding ████
Text ████ CLIP text embedding ██████ 30/70 fusion ██ normalized vector
█
████████████████████████████████████████████████████████████████████████████████
▼ ▼
duplicate detection persona models
█ █
complete-linkage scores + labels
█ █
████████████████████████████████████████████████████████████████████████████████
▼
abstention + evidence
▼
LLM rationale / fallback
▼
JSON output

```

5. Multimodal Representation

5.1 CLIP ViT-B/32 baseline

The primary encoder is openai/clip-vit-base-patch32. The final projected CLIP image and text embeddings are 512-dimensional. Each modality is normalized before fusion, then the fused representation is normalized again.

5.2 Why multimodal fusion

The image captures visual campaign information while the ad text contains explicit semantic and product/audience signals. The project therefore evaluates different image/text contributions rather than assuming either modality should dominate.

5.3 Fusion experiment

Image : Text	Macro-F1	Micro-F1	Weighted-F1	Hamming	Subset
70 : 30	58.00%	59.39%	60.76%	12.64%	34.78%
50 : 50	60.38%	63.81%	64.27%	10.33%	46.74%
30 : 70	61.10%	64.89%	64.13%	8.97%	51.09%
20 : 80	59.54%	66.36%	67.30%	9.78%	46.74%

The selected configuration is 30% image + 70% text. It achieved the highest Macro-F1, lowest Hamming loss, and highest subset accuracy among the tested fusion configurations. The 20:80 configuration had higher Micro-F1 and Weighted-F1, but did not match the selected configuration on Macro-F1 or exact subset accuracy.

5.4 Alternative vision-language model

Google SigLIP Base (ViT-B/16), google/siglip-base-patch16-224, was evaluated using the same leakage-safe split and downstream search. Its final test Macro-F1 was 57.22%, versus 61.10% for CLIP. The project also considered larger CLIP-family candidates, but did not pursue them because the available evidence did not justify the additional model-size and computational cost on this label-limited dataset.

6. Near-Duplicate Detection & Leakage Control

6.1 Similarity-based duplicate detection

Duplicate detection uses cosine similarity between the fused multimodal embeddings. A provisional operating threshold of 0.79 was selected. The 120-pair evaluation set contains 60 duplicate and 60 non-duplicate pairs and was held out for evaluation.

6.2 Full pairwise analysis

Statistic	Value
Creatives	2,882
Unique pairs	4,151,521
Mean similarity	0.51265
90th percentile	0.64643
95th percentile	0.68563
99th percentile	0.75132
99.9th percentile	0.81548
Pairs ≥ 0.79	11,283
Pairs ≥ 0.85	1,341
Pairs ≥ 0.90	513

6.3 Why clustering matters

A major methodological risk is allowing two near-duplicate creatives to appear in different evaluation splits. A model could then appear to generalize while effectively seeing the same campaign during training. The solution is to cluster duplicates first and split at the cluster level.

6.4 Complete-linkage improvement

The first clustering strategy could suffer from chaining: A similar to B and B similar to C could place A, B, and C in one large group even when A and C were not sufficiently similar. The final implementation uses complete-linkage hierarchical clustering at the same 0.79 similarity threshold. This requires all members of a cluster to remain mutually similar.

Measure	Before	After complete linkage
Duplicate edges	11,283	11,283
Clusters	1,923	2,038
Largest cluster	87 creatives	14 creatives

The cluster change is primarily a leakage-control improvement rather than a controlled model-performance improvement. Because changing the clusters changes the train/dev/test composition, score differences across the two cluster definitions should not be interpreted as a pure model gain.

7. Leakage-Safe Train / Dev / Test Design

The final split is performed on duplicate-cluster IDs rather than individual creatives. The target is approximately 70% / 15% / 15% of clusters, while accepting that the creative counts cannot be exactly proportional because entire clusters must remain intact.

Split	Creatives	Clusters
Train	421	343
Development	87	73
Test	92	74

The development set is used for threshold and hyperparameter selection. The test set is used only for the final held-out comparison. This separation is important because repeated experimentation on the test set would turn it into a tuning set.

Selection protocol

- Build duplicate/near-duplicate clusters from the full creative embedding space.
- Assign complete clusters to train, development, or test.
- Train persona models only on training creatives.
- Select model hyperparameters and persona thresholds using development Macro-F1.
- Evaluate the selected configuration once on the held-out test split.

8. Persona Prediction Models

8.1 Initial baseline

The first experiment used the leakage-safe data concept but an earlier modelling setup. It produced relatively weak performance, with kNN Macro-F1 26.25% and Logistic Regression Macro-F1 19.28%. This motivated a redesign around genuine multi-label prediction.

8.2 Multi-label kNN

The improved kNN model uses NearestNeighbors to retrieve similar creatives, then aggregates the persona labels of all retrieved neighbours using similarity-weighted scores. This is preferable to forcing a single class because one creative may legitimately belong to multiple personas.

8.3 One-vs-Rest Logistic Regression

The classifier treats each of the eight personas as an independent binary prediction problem over the fused embedding. This naturally supports multi-label output and allows the model to produce a score for every persona.

8.4 Threshold and hyperparameter search

The final experiment searched kNN values of 3, 5, 7, 9, 11, and 15; Logistic Regression C values from 0.01 through 30; and persona thresholds from 0.20 through 0.70. The best configuration was chosen by development Macro-F1.

Model	Best DEV config	DEV Macro-F1	TEST Macro-F1	TEST Micro-F1	TEST Hamm
kNN	k=5, threshold=0.40	66.40%	49.32%	55.10%	11.96%
One-vs-Rest LR	C=30, threshold=0.60	73.07%	61.10%	64.89%	8.97%

The kNN model achieved strong development performance but dropped substantially on the unseen test clusters. Logistic Regression generalized better in this experiment and was therefore selected as the final persona model.

9. Experimentation & Iteration History

The project did not jump directly to the final configuration. The main improvements were methodological as well as numerical.

Iteration	Change	Reason / observation
1	Initial kNN + classifier comparison	Established a baseline and exposed the need for proper multi-label modelling.
2	Multi-label kNN via weighted neighbour labels	Allowed multiple personas and made retrieved neighbours usable as evidence.
3	Cluster-level train/dev/test split	Prevented duplicate/near-duplicate leakage.
4	Complete-linkage duplicate clustering	Reduced chaining and tightened duplicate groups.
5	Threshold tuning	Moved decision thresholds from arbitrary defaults to development-selected operating points.
6	Hyperparameter tuning	Selected k and Logistic Regression C using development Macro-F1.
7	Image/text fusion search	Established 30:70 as the final multimodal balance.
8	SigLIP comparison	Confirmed that a newer alternative representation did not outperform CLIP on this dataset.
9	Trust-aware inference	Added abstention, evidence retrieval, schema validation, LLM rationale, and determination of final output.

Interpretation of the improvement story

The final gains should not be attributed to a single architectural trick. The strongest methodological improvements came from making the evaluation more faithful to the real problem: multi-label prediction, cluster-level leakage control, development-only selection, and systematic fusion/model comparison.

10. Abstention, Confidence & Evidence

The assignment explicitly treats abstention as a core requirement. The system should not force a persona recommendation when the model has insufficient evidence.

Final operating logic

Condition	Decision	Visible confidence
Top score ≥ 0.60 and top1-top2 margin ≥ 0.10	suggest	high
Otherwise top score ≥ 0.45	suggest_with_caution	medium
Top score < 0.45	abstain	low

The numeric score used here is the classifier's persona score. The visible high/medium/low confidence label is an operating decision derived from that score and the score margin; it should not be interpreted as a separately calibrated probability.

Evidence retrieval

For a selected persona, the system retrieves relevant labelled creatives and reports their creative ID, cosine similarity, and whether the pair is a near-duplicate under the duplicate threshold. Evidence is deduplicated and sorted by similarity.

Why evidence and classifier scores are different

The Logistic Regression score and nearest-neighbour cosine similarity answer different questions. The classifier score comes from the learned decision boundary over the full fused 512-dimensional embedding. Evidence similarity is a pairwise cosine comparison between the query creative and individual reference creatives. Therefore a classifier score can be very high even when no single reference creative is a near-duplicate.

11. Evidence-Constrained LLM Rationale

The LLM is not used to make the persona prediction. The ML model produces the prediction first; the LLM is used only to explain the prediction using supplied evidence.

Prompt boundary

- Predicted persona.
- Model score.
- Retrieved creative IDs.
- Similarity values.
- Near-duplicate flags.
- No full dataset access.

This design is intended to prevent unsupported claims such as inventing objects, demographics, products, or visual themes that are not present in the supplied evidence.

Output validation and fallback

```
LLM → JSON parser → Pydantic / JSON Schema → valid? → use rationale  
■■ no → deterministic fallback
```

The implementation supports a local mock mode (MOCK_LLM=1) for reproducible execution without an API call. In normal mode, the OpenAI API is used for the rationale, while failures fall back to a deterministic local rationale.

Important implementation principle

The explanation layer is intentionally downstream of the predictive model. This keeps the LLM from silently becoming an additional uncontrolled classifier.

12. Inference Interface & Example Behaviour

CLI

```
python -m srcA.cli --creative-id c_0030
python -m srcA.cli --creative-id c_0030 --mock-llm
```

Observed high-confidence example

For creative c_0860, the system predicted Health & Wellness Seeker with a score of approximately 0.952, classified the result as high confidence / suggest, and retrieved multiple supporting examples, including near-duplicate matches.

Observed medium-confidence example

For c_0030, the system predicted Traveler & Leisure with a score of approximately 0.553 and returned suggest_with_caution. This demonstrates that the system does not require every prediction to cross the strong recommendation threshold.

Unconfirmed creative behaviour

An unconfirmed creative such as c_2417 can still be scored by the classifier. Its prediction does not require a direct label lookup in confirmed_personas.csv. The confirmed dataset supplies training labels and reference evidence; the saved classifier produces the prediction for a new creative.

This distinction is important for deployment: the system is a learned predictor, not a lookup table over the 600 confirmed creatives.

13. Repository & Reproducibility Structure

The implementation is separated into focused modules so that data handling, embedding, duplicate detection, modelling, evaluation, trust logic, rationale generation, validation, and CLI inference can be tested or replaced independently.

```
memologs/  
■■■ creatives.parquet  
■■■ confirmed_personas.csv  
■■■ personas.yaml  
■■■ dup_pairs_eval.csv  
■■■ geo_panel.csv  
■■■ images/  
■■■ srcA/  
■ ■■■ config.py  
■ ■■■ data.py  
■ ■■■ embeddings.py  
■ ■■■ duplicate.py  
■ ■■■ clustering.py  
■ ■■■ split.py  
■ ■■■ knn_model.py  
■ ■■■ classifier.py  
■ ■■■ evaluate.py  
■ ■■■ abstention.py  
■ ■■■ evidence.py  
■ ■■■ rationale.py  
■ ■■■ schema.py  
■ ■■■ pipeline.py  
■ ■■■ cli.py  
■■■ scripts/  
■■■ artifacts/  
■■■ REPORT.md  
■■■ README.md  
■■■ Makefile  
■■■ pyproject.toml
```

Reproducibility principles

- Fixed random seed used in model training/evaluation.
- Cached embedding/artifact paths separated from source data.
- Development set controls threshold and hyperparameter selection.
- Held-out test set remains separate from tuning decisions.
- Mock LLM mode supports deterministic local pipeline execution.

14. Limitations & Further Improvements

Current limitations

- The labelled dataset contains only 600 creatives and is strongly imbalanced across personas.
- Classifier scores are operating scores rather than formally calibrated probabilities.
- The current evidence payload is primarily IDs, similarities, labels, and duplicate flags; it does not directly provide rich textual or visual evidence to the rationale model.
- The duplicate threshold of 0.79 is a chosen operating point; a dedicated development set for duplicate-threshold selection would make this choice more principled.
- The report's final model-selection results are tied to this dataset and split composition; performance should not be assumed to transfer unchanged to a different ad inventory.

Potential next steps

- Calibrate persona probabilities using a leakage-safe calibration procedure.
- Add per-persona precision/recall/F1 and confidence calibration analysis, especially for minority personas.
- Create a dedicated duplicate threshold-development protocol and report precision/recall/F1 explicitly on the held-out pair set.
- Enrich evidence passed to the rationale layer with controlled snippets or modality-level evidence, while retaining strict grounding constraints.
- Monitor abstention coverage versus precision after deployment.
- Evaluate temporal or campaign-level drift if new creatives arrive continuously.

15. Final Results & Conclusion

Final selected configuration

Component	Final choice
Image/text encoder	OpenAI CLIP ViT-B/32
Embedding dimension	512
Fusion	30% image + 70% text
Duplicate similarity threshold	0.79
Duplicate clustering	Complete-linkage hierarchical clustering
Split	421 train / 87 dev / 92 test creatives
Persona baseline	Weighted multi-label kNN
Final persona model	One-vs-Rest Logistic Regression
LR hyperparameter	$C = 30$
Persona decision threshold	0.60
Abstention strong threshold	$0.60 + \text{margin} \geq 0.10$
Abstention caution threshold	0.45
Rationale	Evidence-constrained LLM + deterministic fallback

Final held-out persona metrics

Metric	kNN	Final Logistic Regression
Macro-F1	49.32%	61.10%
Micro-F1	55.10%	64.89%
Weighted-F1	56.05%	64.13%
Hamming loss	11.96%	8.97%
Subset accuracy	45.65%	51.09%

Part A therefore ends with a lightweight but structured ML system: pretrained multimodal representation learning, explicit duplicate controls, leakage-safe evaluation, multi-label prediction, abstention, retrieved evidence, and grounded explanation. The final result is not simply a classifier score; it is a complete inference workflow designed around the assignment's central question: "Should I trust this recommendation?"

Source note

All numerical results, dataset statistics, experiment descriptions, architectural choices, and example behaviours in this report were compiled from the project notes supplied for the assignment.